insurmountable inefficiency with an order-book matching is that both counterparties must be willing and able to exchange at the agreed on rate for the trade to execute. This requirement creates limitations for many smart contract applications in which demand for exchange liquidity cannot be dependent on a counterparty's availability. An innovative alternative is an AMM.

Automated Market Makers

An AMM is a smart contract that holds assets on both sides of a trading pair and continuously quotes a price for buying and for selling. Based on executed purchases and sales, the contract updates the asset size behind the bid and the ask and uses this ratio to define its pricing function. The contract can also account for more complex data than relative bid–ask size when determining price. From the contract's perspective, the price should be risk-neutral where it is indifferent to buying or selling.

A naive AMM might set a fixed price ratio between two assets. With a fixed price ratio, when the market price shifts between the assets, the more valuable asset would be drained from the AMM and arbitraged on another exchange where trading is occurring at the market price. The AMM should have a pricing function that can converge to an asset's market price. That is, the pricing function makes it more expensive to purchase the asset from the trading pair as the ratio of the asset to others in the contract decreases.